GATE-2008

BT: BIOTECHNOLOGY

Q. 1 – 6 carry one mark

1. Diauxic pattern of biomass growth is associated with

(P) multiple lag phases

(Q) sequential utilization of multiple substrates

(R) simultaneous utilization of multiple substrates

(S) absence of lag phase

(a) P, R (b) P, Q

(c) R, S (d) Q, S

2. Zinc fingers are characteristics of

(a) blood clotting proteins

(b) RNA chaperones

(c) DNA binding proteins

(d) lysosomal hydrolases

3. Parthenogenetic embryos in plants are those which are formed by

(a) unfertilized eggs

(b) fertilized eggs

(c) sporophytic cells

(d) male gametophyte

4. Which one of the following is the growth factor used for growth of tissues and organs in plant tissue culture?

(a) Cysteine

(b) Cytokinin

(c) Cytidylate

(d) Cyclic AMP

5. Which of the following techniques is best suited for immobilizing an affinity ligand?

(a) Physical adsorption

(b) Gel entrapment

(c) Cross-linking with a polymer

(d) Covalent linkage to a spacer arm

6. Multiplication of genetically identical copies of a cultivar by asexual reproduction is known as

(a) aclonal propagation

(b) vegetative propagation

(c) polyclonal propagation

(d) clonal propagation

Q. 7 – 24 carry two marks each.

7. Identify the correct statement for the 'HAT medium'

(P) Includes drug aminopterin to block major pathway for synthesis of deoxyri-bonucleotides

(Q) Hypoxanthine is precursor for thymidine

(R) Includes drug aminopterin to block major pathway for synthesis of polypeptides

(S) Cells can grow in presence of aminopterin only if they have enzymes thymidine kinase and hypoxanthine-guanine phosphoribosyl transferase

(a) P, Q (b) P, S

(c) R, S (d) Q, S

8. A DNA fragment of 4500 bp has to be tailed with dT residues by using dTTP and the enzyme 'terminal transferase'. The stock solution of dTTP that is used as a substrate has a concentration of 150μM. Ten μl of this stock solution is added to a total volume of 200μl reaction. What will be the concentration of dTTP in the reaction?

(a) 7.5 μM (b) 75 μM

(c) 0.75 μM (d) 0.075 μM

9. Determine the correctness or otherwise of following **Assertion [a] and Reason [r]**

Assertion: The enzymatic degradation of cell wall to obtain single cell called protoplast has helped immensely in developing somatic cell genetics in plants

Reason: In plants or animals, fusion to two cells must occur through the plasma membrane

(a) Both [a] and [r] are true and [r] is the correct reason for [a]

(b) Both [a] and [r] are true but [r] is not the correct reason for [a]

(c) [a] is true but [r] is false

(d) [a] is false but [r] is true

10. In bioinformatics, the term 'BLAST' refers to

(a) database retrieval tool

(b) computational tool for sequence homology searching and alignment

(c) computational tool to view genomic sequences

(d) computational tool to view protein structures

11. Match the terms in group 1 with their possible explanations in group 2

Group 1	Group 2
P. Orthologs	1. A cell or an organsim having foreign gene
Q. Paralogs	2. The complement of a protein expressed by a genome
R. Proteome	3. Genes from different spe-cies related to each other
S. Transgenic	4. Genes from same species related to each other

	P	Q	R	S
(a)	2	4	1	3
(b)	4	3	2	1
(c)	3	4	2	1
(d)	1	2	3	4

12. Which of the following statements are true with respect to a special complex called 'dicer'?

(P) It consists of deoxyribonuclease and DNA fragments

(Q) It consists of ribonuclease and RNA fragments

(R) It is involved in gene silencing

(S) It triggers apoptosis

(a) P, R (b) Q, R

(c) P, S (d) Q, S

13. Some living cells (e.g. plant cell) have the capacity to give rise to whole organism. The term used to describe this property is

(a) morphogenesis (b) androgenesis

(c) totipotency (d) organogenesis

14. Match the items in groupl with the terms given in group 2

Group 1	Group 2
P. *Lactobacillus and Bifidobacteria*	1. Prebiotics
Q. Polychlorobenzenes (PCBs)	2. Probiotics
R. Fructo-oligosaccharides	3. Antibiotics
S. β-Lactams	4. Xenobiotics

	P	Q	R	S
(a)	2	4	1	3
(b)	3	4	1	2
(c)	4	1	2	3
(d)	1	3	4	2

15. Match the coefficients in group 1 with their corresponding downstream processing steps given in group 2

Group 1	Group 2
P. Sedimentation coefficient	1. Aqueous two phase extraction
Q. Partition coefficient	2. Ultrafiltration
R. Rejection coefficient	3. Dialysis
S. Activity coefficient	4. Centrifugation

	P	Q	R	S
(a)	3	1	4	2
(b)	2	1	4	3
(c)	4	3	1	2
(d)	4	1	2	3

16. Match the bioreactor components in group 1 with the most appropriate function given in group 2

Group 1	Group 2
P. Marine type impeller	1. Recirculation of medium
Q. Draft tube	2. Aeration of medium
R. Diaphragm valve	3. Animal cell cultivation
S. Sparger	4. Sterile operation

	P	Q	R	S
(a)	4	2	1	3
(b)	3	1	4	2
(c)	3	4	2	1
(d)	2	1	4	3

17. Evaluate the Michaelis constant for the following lipase catalyzed trans-esterification reaction for the production of biodiesel

$$\text{Vegetable oil} + \text{Lipase} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \text{Oil-lipase complex} \xrightarrow{k_3} \text{Biodiesel} + \text{Glycerol}$$

where, $k_1 = 3 \times 10^8\ M^{-1}\ s^{-1}$; $k = 4 \times 10^3\ s^{-1}$ and $k_2 = 2 \times 10^3\ s^{-1}$

(a) 4.2×10^{-3} M (b) 14.0×10^{-4} M

(c) 6.4×10^{-6} M (d) 1.4×10^{-4} M

18. In a chemostat, evaluate the dilution rate at the cell wash-out condition by applying Monod's model with the given set of data: $\mu_{max} = 1 h^{-1}$; $Y_{x/s} = 0.5$ g g^{-1}; $K_s = 0.2$ g L^{-1}; $S_0 = 10$ g L^{-1}

(a) 1.00 h^{-1} (b) 0.49 h^{-1}

(c) 0.98 h^{-1} (d) 1.02 h^{-1}

19. Match the products in group 1 with their producer organisms given in group 2

Group 1	Group 2
P. Ethanol	1. *Streptomyces orientalis*
Q. L-Lysine	2. *Saccharomyces cerevisiae*
R. Biopesticide	3. *Corynebacterium glutamicum*
S. Vancomycin	4. *Bacillus thuringiensis*

	P	Q	R	S
(a)	2	3	4	1
(b)	3	4	1	2
(c)	4	1	2	3
(d)	2	1	4	3

20. A polymerase chain reaction was performed beginning with 400 template DNA molecules in a 100 µl reaction. After 20 cycles of polymerase chain reaction, how many molecules of the amplified product will be present in O.1µl of reaction ?

(a) 2.19×10^4 (b) 4.19×10^4

(c) 2.19×10^5 (d) 4.19×10^5

21. A bacterial culture with an approximate biomass composition of $CH_{1.8}O_{0.5}N_{0.2}$ is grown aerobically on a defined medium containing glucose as the sole carbon source and ammonia being the nitrogen source. In this fermentation, biomass is formed with a yield coefficient of 0.35 gram dry cell weight per gram of glucose and acetate is produced with a yield coefficient of 0.1 gram acetate per gram of glucose. The respiratory coefficient for the above culture will be

(a) 0.90 (b) 0.95

(c) 1.00 (d) 1.05

22. A bacterial culture having a specific oxygen uptake rate of 5 mmol O_2 $(g\text{-}DCW)^{-1}$ hr^{-1} is being grown aerobically in a fed-batch bioreactor. The maximum value of the volumetric oxygen transfer coefficient is $0.18s^{-1}$ for the stirred tank bioreactor and the critical dissolved oxygen concentration is 20% of the saturation concentration (8mg/ml). The maximum density to which the cells can be grown in the fed-batch process without the growth being limited by oxygen transfer, is approximately.

(a) 14 g/l (b) 26 g/l

(c) 32 g/l (d) 65 g/l

Common Data Questions

Common Data for Questions 23 and 24:

An enzyme (24000 Da) undergoes first-order deactivation kinetics while catalyzing a reaction according to Michaelis-Menten kinetics ($K_m = 10^{-4}$ M). The enzyme has a turnover number if 10^4 molecules substrate/min-(molecule enzyme) and a deactivation constant (k_d) of 0.1 min^{-1} at the reaction conditions. The reaction mixture initially contains 0.6 mg/l of active enzyme and 0.02 M of the substrate.

23. The time required to convert 10% of the substrate will be rapproximately

(a) 16 min (b) 24 min

(c) 32 min (d) 8 min

24. The maximum possible conversion for the enzymatic reaction will be

(a) 100% (b) 50%

(c) 25% (d) 12.5%

Linked Answer Questions: Q.25 to Q.28 carry two marks each.

Statement for Linked Answer Questions 25 and 26:

A Nick Translation reaction in a final volume of 100µl was carried out by using 25 µCi of labeled [α-^{32}P]–dCTP for labeling a 1.2 Kb λ-Interferon DNA fragment.

25. After completion of Nick translation reaction , 10 µl of reaction was spotted on a glass-fiber filter that upon counting resulted into 4.2×10^4 cpm in reaction. Another 10 µl was processed for TCA precipitation to determine redioisotope incorporation. The TCA precipitated sample gave 2.94×10^4 cpm. What is the percent of [α-^{32}P]-dCTP incorporation into the DNA sample?

(a) 40% (b) 50%

(c) 60% (d) 70%

26. If 2.94×10^4 cpm of TCA precipitable counts of the 10 µl sample were taken from 1/10 dilutioa of the 100µl Nick Translation reaction containing 1 µg of γ-Interferon DNA, what is the specific activity of the labeled product?

(a) 1.47×10^6 cpm/µg

(b) 1.47×10^7 cpm/µg

(c) 2.94×10^6 cpm/µg

(d) 2.94×10^7 cpm/µg

Statement for Linked Answer Questions 27 and 28:

A double reciprocal plot was created from the specific growth rate and limiting-substrate concentration data obtained from a chemostat experiment. A linear regression gave values of 1.25 hr and 100 mg-hr-l^{-1} for the intercept and slope, respectively.

27. The respective values of the Monod kinetic constants μ_m (hr^{-1}) and K_s (mg/l) are as follows:

(a) 0.08, 8 (b) 0.8, 0.8

(c) 0.8, 80 (d) 8, 8

28. The same culture (with the μ_m and K_s values as computed above) is cultivated in a 10-litre chemostat being operated with a 50 ml/min sterile feed containing 50 g/l of substrate. Assuming an overall yield coefficient of 0.3 g-DCW/g-substrate, the respective values of the outlet biomass and substrate concentration are

(a) 15 g/l, 48 mg/l (b) 15 g/l, 0.48 g/l

(c) 48 g/l, 15 g/l (b) 4.8 g/l, 4.8 g/l

ANSWERS

1. (b)	2. (c)	3. (a)	4. (b)	5. (d)	6. (d)	7. (b)	8. (a)	9. (b)	10. (b)
11. (c)	12. (b)	13. (c)	14. (a)	15. (d)	16. (b)	17. (d)	18. (c)	19. (a)	20. (d)
21. (b)	22. (d)	23. (*)	24. (*)	25. (d)	26. (d)	27. (c)	28. (a)		

EXPLANATIONS

1. When both the substrates are present in a batch bioreactor, microbial cells do not consume both substrates simultaneously or additively but instead grow on the sugars sequentially. The sequential consumption results in an interesting pattern of the cells first consuming one preferred substrate and then after an intermediate lag phase, consuming the remaining less preferred substrate. That is, the less preferred substrate is not utilized until the preferred substrate is completely consumed and is no longer available in the growth medium. This sequential utilization of two substrates in batch cultures has been observed in numerous experiments by Monod, who has termed this phenomenon as the "diauxic pattern of growth". It has multiple lag phases.

2. Zinc fingers function as interaction modules that bind DNA, RNA, proteins, or other small, useful molecules.

3. Parthenogenesis is a form of asexual reproduction in which growth and development of embryos occur without fertilization. In plants, parthenogenesis means development of an embryo from an unfertilized egg cell, and is a component process of apomixis.

4. Cytokinins (CK) are a class of plant growth substances (phytohormones) that promote cell division, or cytokinesis, in plant roots and shoots. They are involved primarily in cell growth and differentiation, but also affect apical dominance, axillary bud growth, and leaf senescence. Folke Skoog discovered their effects using coconut milk in the 1940s at the University of Wisconsin-Madison.

5. Affinity chromatography utilizes the specific interactions between two molecules for the purification of a target molecule. A ligand having affinity for a target molecule is covalently attached to an insoluble support and functions as bait for capturing the target from complex solutions. The affinity ligand can be any molecule that will bind the target without also binding other molecules in the solution.

6. By clonal propogation or micropropagation thousands of plants can be produced from a single piece of plant tissue explant. They are genetically identical.

7. STATEMENT P: HAT Medium (hypoxanthine-aminopterin-thymidine medium) is a selection medium for mammalian cell culture, which relies on the combination of aminopterin, a drug that acts as a powerful folate metabolism inhibitor by inhibiting dihydrofolate reductase, with hypoxanthine and thymidine which are intermediates in DNA synthesis. Aminopterin blocks DNA de novo synthesis.

   STATEMENT S: The thymidine present in the HAT medium that can be absorbed by the cells and phosphorylated by thymidine kinase (TK) into TMP. By additional phosphorylation reactions, TMP can be used to make thymidine triphosphate (TTP), one of the four nucleotide precursors that are used by DNA polymerases to create DNA. Hypoxanthine-guanine phosphoribosyltransferase (HGPRT) reacts with hypoxanthine absorbed from the medium with PRPP, liberating pyrophosphate, to produce IMP(Inosine Mono Phosphate) by a salvage pathway.

   Therefore, the use of HAT medium for cell culture is a form of artificial selection for cells containing working TK and HGPRT.

8. Stock solution= 150 µM

   Dilution factor= 10/200 µl = 1/20 µl

   Concentration of d TTP= (1/20)*150 µM= 7.5 µM

9. Somatic fusion, also called protoplast fusion through plasma membrane, is a type of genetic modification in plants by which two distinct species of plants are fused together to form a new hybrid plant with the characteristics of both, a somatic hybrid. Hybrids have been produced either between the different varieties of the same species (e.g. between non-flowering potato plants and flowering potato plants) or between two different species (e.g. between wheat triticum and rye secale to produce Triticale).

10. In bioinformatics, Basic Local Alignment Search Tool, or BLAST, is an algorithm for comparing primary biological sequence information, such as

the amino-acid sequences of different proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold.

11. Orthologs, or orthologous genes, are genes in different species that originated by vertical descent from a single gene of the last common ancestor. Paralogous sequences provide useful insight into the way genomes evolve. The genes encoding myoglobin and hemoglobin are considered to be ancient paralogs.

12. • Dicer is an endoribonuclease in the RNase III family that cleaves double-stranded RNA (dsRNA) and pre-microRNA (miRNA) into short double-stranded RNA fragments about 20-25 base pairs long, with a two-base overhang on the 3' end. Dicer contains two RNase III domains and one PAZ domain; the distance between these two regions of the molecule is determined by the length and angle of the connector helix and influences the length of the siRNAs it produces.

• Dicer facilitates the formation of the RNA-induced silencing complex (RISC), whose catalytic component argonaute is an endonuclease capable of degrading messenger RNA (mRNA).

13. Totipotency is the ability of a single cell to divide and produce all the differentiated cells in an organism, including extraembryonic tissues. Totipotent cells include spores and zygotes.

14. • **Probiotics** are live bacteria that may confer a health benefit on the host. Lactic acid bacteria (LAB) and bifidobacteria are the most common types of microbes used as probiotics.

• A **xenobiotic** is a chemical which is found in an organism but which is not normally produced or expected to be present in it. The term xenobiotics is very often used in the context of pollutants such as dioxins and polychlorinated biphenyls.

• **Fructo-oligosaccharide** (FOS) and **lactulose** have been classified as prebiotics. Prebiotics are non-digestible food ingredients that stimulate the growth and/or activity of bacteria in the digestive system in ways claimed to be beneficial to health.

• **β-Lactam antibiotics** (beta-lactam antibiotics) are a broad class of antibiotics, consisting of all antibiotic agents that contains a β-lactam ring in their molecular structures.

15. Svedbarg Unit stands for sedimentation coefficient for centrifugation process. Partition coefficient for two layer aqueous extraction unit. Rejection coefficient is the unit for ultra filtration. Dialysis unit is activity coefficient.

16. Sparger is being used for the aeration of the medium in bioreactor. Diaphram valve is being used for the air sterilization in large scale animal or microbial cell culture condition. Draft tube has been used for the feed circulation in semi fedbatch culture. Marine type turbine is being used to reduce shear stress on fluid medium in animal cell culture.

17. $K_M = \dfrac{K_{-1} + K_2}{K_1}$

$$= \frac{4 \times 10^4\ S^{-1} + 2 \times 10^3\ S^{-1}}{3 \times 10^8\ M^{-1} S^{-1}}$$

$$= \frac{42000}{3 \times 10^8}$$

$$= 1.4 \times 10^{-4} M$$

18. We know, from Monod's equation

$$\mu = \mu_{max} [S_0/(K_S + S_0)]$$

$$\Rightarrow \mu = 1 * [10/(0.2 + 10)] = 10/10.2 = 0.98$$

At wash out, Dilution rate $D = \mu$

Therefore,

the required dilution rate = 0.98 hr^{-1}

19. **P>2:** Saccharomyces cerevisiae is used extensively in batch fermentations to convert sugars to ethanol for the production of beverages and biofuels.

Q>3: Industrially, L-lysine is usually manufactured by a fermentation process using Corynebacterium glutamicum

R>4: Bacillus thuringiensis (or Bt) is a Gram-positive, soil-dwelling bacterium, commonly used as a biological pesticide.

S>1: Streptomyces orientalis produces the antibiotic Vancomycin, which is used for the prevention and treatment of infections caused by Gram-positive bacteria, especially staphylococci.

20. In PCR, the template DNA after 'n' cycle

$$= (\text{original DNA}) \times 2^n$$

$$= 400 \times 2^{20}$$

But, this is amount in 100μl. In 0.1 μl we have,

$$= \frac{400 \times 2^{20} \times 0.1}{100}$$

$$= 4.194304 \times 10^5$$

21. $Y_{x/s} = 0.35$ g/g, $Y_{P/S} = 0.1 \frac{\text{g acetate}}{\text{g glucose}} \frac{\text{g dry cell}}{\text{g glucose}}$

The respiratory coefficient

$$\text{Respiratory quotient} = \frac{\text{Moles of } CO_2 \text{ formed}}{\text{Moles of } CO_2 \text{ consumed}}$$

$$= \frac{5}{4.95} = 1.01$$

$$C_6H_{12}O_6 + bNH_3 + O_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + eH_2O + fCO_2$$

C : $6a = 1 + f$

H : $12a + 3b = 1.8 + 2e$ ...(1)

O : $6a + 2d = 0.5 + e + 2F$ ...(2)

$b = 0.2$ ...(3)

N : $b = 0.2$

$\therefore$ $12a = 1.2 + 2e$

$6a = 1 + F$

Let if $a = 1$, i.e. $F = 5$

$e = 5.4$

$d = 4.95$

$b = 0.2$

$a = 1$

22. $K_La = 0.185\ S^{-1} = 5.139 \times 10^{-5}\ hr^{-1}$

Critical dO_2 is 20% of the saturate conc.(8 mg/ml)

Cell density. $C_L = 1.6 \times 10^3$.m/l $C = (8 \times 10^3$ mg/l)

$$do_2 = K_La\ (C - C_L)$$

$$OUR = \frac{1/\mu}{1/o_2}$$

$$= q_{o2}X.$$

q_{o2} = mmo2/gm sw. hr.

$= 5 \times 10^{-3}$ gm dm hr

$\therefore K_{La}(C - C_L) = q_{02}X.$

$$\Rightarrow \quad X = \frac{K_{La}(C - C_L)}{q_{02}}$$

$$\therefore \quad X = \frac{5.139 \times 10^{-5}(8 - 1.6) \times 10^3}{5 \times 10^{-3}}$$

$= 65.779$ g/lit.

25. % incorporation is calculated by incorporation int precipitated form. (The polymerized form wil precipitate).

$$= \frac{\text{precipitate form cpm}}{\text{total cpm}}$$

$$= \frac{2.94 \times 10^4 \text{ cpm}}{4.2 \times 10^4 \text{ cpm}} \times 100 \cong 70\%$$

26. 2.94×10^5 cpm is from 0.1 dilution of 10 µl sample. Thus, 10 µl sample will have 2.94×10^6 cpm but this 10 µl sample is taken from 100 µl which contain 1 µg of λ-interferon. So, 100 µl of 1 µg interferon sample will have 2.94×10^7 cmp/µg.